

SHREYA BALAKRISHNA

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EDUCATION

University of Massachusetts Amherst

Master of Science in Computer Science, GPA: 3.789/4

Amherst, MA

Sep. 2024 – May. 2026

Coursework: Theoretical Machine Learning, Advanced NLP, Information Retrieval, Advanced Algorithms

Vellore Institute of Technology

Bachelor of Technology in Computer Science and Engineering, GPA: 8.71/10

Tamil Nadu, India

Sep. 2020 – Jul. 2024

Coursework: Computer Vision, DBMS, DSA, Computational Social Science, Cloud Computing, Operating Systems

TECHNICAL SKILLS

ML & AI: PyTorch, TensorFlow, vLLM, LoRA, HF Accelerate, HF Transformers, Supervised Fine-Tuning, Weaviate, ChromaDB, Vector Databases, LangChain, LangGraph, OpenAI API, RAG, RLHF, Reward Modeling, Weights & Biases

Data & Analytics: NumPy, Pandas, Scikit-Learn, Matplotlib, SQL, BigQuery, ETL, PySpark

Frameworks & Tools: FastAPI, CI/CD, HPC, Flask, Django, React, Node.js, Docker, Slurm, Git, MongoDB, WebSocket, REST, AWS

Languages: Python, C++, JavaScript, TypeScript, Bash/Shell, R

Certifications: Google Data Analytics Professional (Coursera), AI Analyst (IBM), Data Analytics Using Big Data (AWS, NUS)

EXPERIENCE

Research Extern

Amherst, MA

Oracle

Jan. 2026 – Present

- Developed a gradient-free LLM training framework combining **Evolution Strategies** with latent chain-of-thought reasoning, enabling **backpropagation-free** CoT training on **Qwen2.5-3B** with zero intermediate activation storage.
- Designed and benchmarked 3 gradient-free latent architectures (weighted softmax on top-k tokens, residual MLP head, cross-attention head) within the ES training loop.
- Achieved 34.96% test reward on **Weighted Softmax** method on top-10 tokens vs. 24.18% on ES baseline, a **44% relative gain**, outperforming **vanilla COCONUT** (20.00%), validating token-embedding-grounded representations over hidden-state injection.

Graduate Researcher

Amherst, MA

Center for Intelligent Information Retrieval

Jul. 2025 – Jan. 2026

- Curated a domain-specific **benchmark** of 500K+ WildChat dialogues filtered by repetition density to study **redundant token generation** in LLM outputs, enabling reproducible measurement of copy behavior across settings.
- Fine-tuned **Llama-3.2-1B-Instruct** to detect and mark redundant content during generation using a custom structured syntax tag, achieving 75.1% train and 71.3% eval accuracy on syntax-compliant generation with a **<4% generalization gap**.
- Validated the reuse-tagging pipeline across **DeepSeek V3**, **Qwen 3**, and **Kimi K2** under varying repetition conditions, confirming **longest-common-substring** based detection generalizes across model families.

Data Science Intern

Mumbai, India

Metropolis Healthcare

Jan. 2024 – May 2024

- Deployed a production **summarization pipeline**: **T5** for extractive compression and **GPT-3.5** for language generation, with a **RAG layer** grounding outputs in patient-specific clinical records; served via **AI chatbot** with 20% improvement in end-user satisfaction.
- Eliminated 12+ hours of manual weekly reporting by building **ETL pipelines** over 2M+ rows and automating **Tableau & SQL** KPI dashboards on **AWS S3**.

Research Intern

Singapore, Singapore

National University of Singapore

Jun. 2023 – Jul. 2023

- Built SAFESCAN, a **YOLOv8**-based PPE detection system achieving 82% accuracy at **30 FPS** on live video.
- Optimized inference pipeline to maintain **real-time throughput** without sacrificing detection precision; deployed via **Flask** for multi-class monitoring.

Data Analyst Intern

Bangalore, India

Stockone Technologies (Acquired by Shopsy)

Apr. 2023 – Jun. 2023

- Queried 5M+ row **PostgreSQL** backend for warehouse KPI dashboards (**Power BI**) across 3 warehouses; extended **Django** auth with **Sentry**-powered real-time session alerting.

PROJECTS

EI Assistant via Hybrid RAG+CAG Architecture | Python, HF Transformers, ChromaDB, Evaluation

- Built a multi-turn emotional intelligence assistant by fine-tuning GPT-4o-mini and LLaMA 3.2-1B with a hybrid RAG+CAG pipeline: RAG for session memory via ChromaDB, CAG for therapeutic guidelines.
- Evaluated fine-tuned models against a held-out test set: 67.5% semantic similarity (vs. 63% pre-fine-tune baseline); human evaluation by a Licensed Clinical Psychologist averaged 4.5/5 across empathy, clarity and supportiveness.

Realtime Chat Platform (ongoing) | FastAPI, Docker, Next.js, TypeScript, PostgreSQL, Redis, WebSocket, pgvector

- Building a horizontally-scalable real-time chat platform with AI-assisted threads (RAG over message history), semantic search via pgvector embeddings, and presence/typing indicators over Redis Pub/Sub.
- Targeting sub-100ms P99 message latency at 10K concurrent WebSocket connections; shipping behind Docker + GitHub Actions CI/CD with structured logging and Prometheus metrics.